

Residual-shape inference and external-proxy audit of structural disturbance in SPARC rotation curves

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Abstract

We ask whether fixed rotation-curve residual-shape features separate externally reviewed structural-disturbance labels in SPARC galaxies, and whether external HI disturbance proxies support the same diagnostic direction. The study reverses the Paper 1 audit: the A/C labels are treated as frozen external targets, while residual features are evaluated under leave-one-galaxy-out thresholding, shuffled-label null tests, bootstrap uncertainty, baseline-family controls, and observability stress checks. The primary internal feature, Projection RMS, reaches LOOGO AUC=0.771008403 with shuffled-label $p = 0.002000000$ and bootstrap 95% AUC interval [0.600802469, 0.909100262]. MOND-simple and empirical RAR-like residual scores also separate A/C systems, whereas a Newtonian baryonic RMS control is near chance. These numbers establish marginal or unconditional class separation under the fixed rule, not projection-formula uniqueness or conditional information beyond shared covariates and residual structure. A later cross-fitted Paper 3 audit found no stable projection-contrast predictive increment, and a source-frozen seven-galaxy EDGE-CALIFA morphology-proxy stress test failed its directional endpoint. The original internal statistics remain unchanged, but the current claim is an unconditional within-SPARC association with mixed external context, not a Tau Core validation claim, gravity-model selection, projection-specific conditional information, or independent paper-grade external validation.

1 Introduction

Rotation-curve residuals are usually discussed as model error, but their radial structure can also be associated with non-equilibrium dynamics, non-circular motions, pressure support, beam smearing, inclination, and source-dependent observability [11, 9]. Paper 1 established a residual-blind marginal association between externally assigned structural-disturbance labels and low-acceleration residual scatter in SPARC. Here we ask the inverse diagnostic question: do fixed residual-shape features separate those external labels better than chance in the same frozen packet?

The scope is deliberately narrow. This paper does not use residuals to redefine the labels, does not validate Tau Core, and does not select a unique gravity law. It quantifies marginal discrimination by a frozen residual feature map and then checks whether external HI disturbance proxies point in the same direction. The original design does not by itself test conditional predictive increment beyond observability, baryonic structure, or low-acceleration residual structure shared with MOND/RAR-like baselines.

2 Data and frozen inputs

The working A/C sample contains 45 SPARC galaxies inherited from the Paper 1 reproducibility packet: 17 externally regular A systems and 28 externally disturbed C systems. SPARC rotation-curve and mass-model context follows [5, 4]. B-class systems remain excluded from the primary target labels because they encode uncertainty by construction. The residual features are computed from the fixed Paper 1 point map and are not retrained on the external-proxy catalogues.

The full reproducibility package, including the frozen derived tables, baseline-score comparisons, control summaries, figures, arXiv source package, and regeneration script, is archived at doi:10.5281/zenodo.20285862. The analysis can be regenerated with the commands listed in the repository README. Raw survey products and raw SPARC rotmod files are not redistributed by this repository.

The projection-family score is treated operationally as a fixed residual map inherited from Paper 1, without requiring the physical correctness of the underlying projection ansatz. This convention is central to the audit: the paper tests marginal residual-label discrimination, not the physical validity of the projection model or a projection-specific conditional-information claim.

3 Residual-shape endpoint and validation design

For each galaxy g and radial point i , the packet starts from absolute log-residual magnitudes

$$r_{m,gi} = |\log V_{\text{obs},gi} - \log V_{m,gi}|,$$

where m denotes the fixed residual family. Projection RMS is then

$$\text{RMS}_{m,g} = \sqrt{\frac{1}{N_g} \sum_i r_{m,gi}^2}.$$

All points are weighted equally; no velocity-error weighting is applied in the primary endpoint. The radial domain is the available SPARC rotation-curve support for each galaxy after the inherited Paper 1 point-map construction. The same aggregation is used for Projection RMS, MOND-simple RMS, RAR-like RMS, and Newtonian baryonic RMS. The Projection endpoint is used as an operational residual-shape score rather than as a physical proof of the projection formula.

The primary internal validation uses leave-one-galaxy-out thresholding. For held-out galaxy h , the threshold is recomputed from the remaining galaxies:

$$T_h = \frac{1}{2} [\text{median}(\text{RMS}_{g \in A, g \neq h}) + \text{median}(\text{RMS}_{g \in C, g \neq h})].$$

The held-out galaxy is classified as C-like if its score exceeds T_h . A shuffled-label null preserves the 17/28 A/C class counts and uses 1000 shuffles with random seed 20260515; the practical p-value floor is therefore of order 10^{-3} , and the observed empirical value is $p = 0.002000000$. Bootstrap uncertainty uses 2000 class-stratified galaxy resamples with random seed 20260516 for the primary endpoint. These tests are internal to SPARC and should not be described as independent validation.

4 Primary results

Table 1: Primary residual-inference results.

Quantity	Value	Interpretation	Claim boundary
Projection RMS LOOGO AUC	0.771008403	primary diagnostic	SPARC-internal
Projection RMS shuffled-label p	0.002000000	label-null check	not replication
MOND-simple RMS AUC	0.720588235	low-acceleration family	not unique
RAR-like RMS AUC	0.731092437	low-acceleration family	not model selection
Newtonian baryonic RMS AUC	0.506302521	near-chance control	limited inference

Table 2: Bootstrap AUC intervals for baseline residual-score families.

Score family	AUC	CI low	CI high
Projection RMS	0.771008403	0.609243697	0.913865546
MOND-simple RMS	0.720588235	0.552521008	0.871848739
RAR-like RMS	0.731092437	0.569275210	0.878151261
Newtonian baryonic RMS	0.506302521	0.340336134	0.678571429

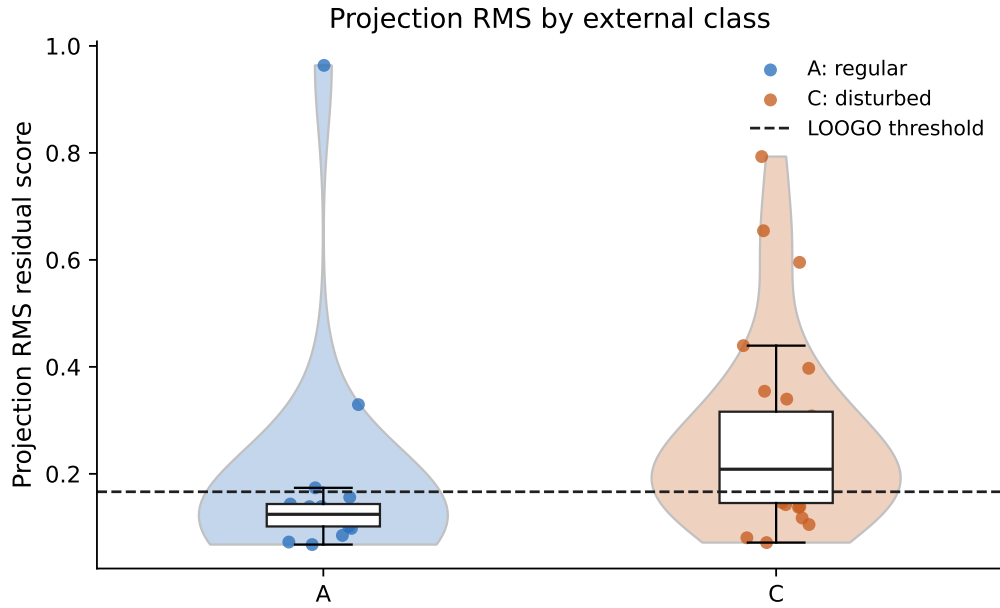


Figure 1: Distribution of the fixed Projection RMS residual score for externally regular A and disturbed C systems.

The primary historical signal is positive as a univariate/rank-discrimination result. Projection RMS separates the frozen A/C labels above chance, and the shuffled-label null places the observed AUC in the extreme tail of the internal null distribution. The bootstrap interval is broad because the sample is small. The correct reading is therefore sample-limited marginal SPARC-internal class

separation under this feature, not evidence that the feature adds conditional information beyond nuisance covariates or common low-acceleration residual structure, and not a physical detection of an underlying field.

Classifier success/failure rotation-curve context

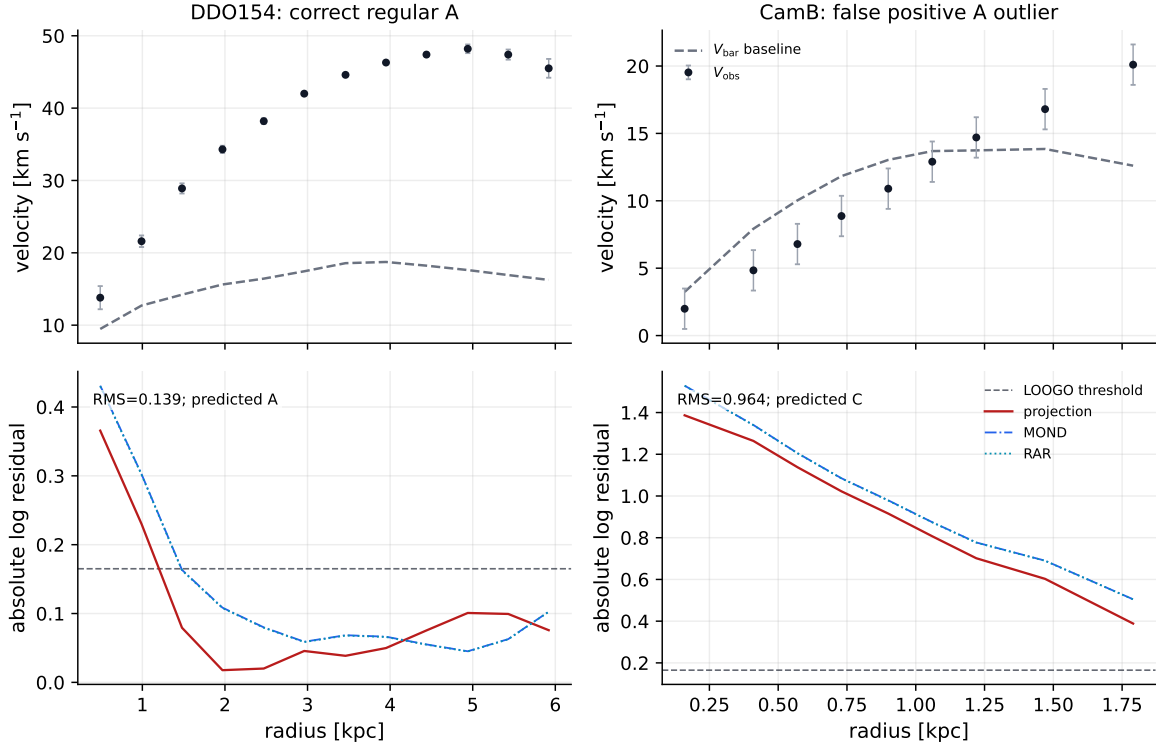


Figure 2: Classifier-context rotation-curve diagnostics for DDO154, a correctly recovered regular A system, and CamB, the named false-positive A outlier. The upper panels show observed rotation curves against the baryonic baseline; the lower panels show per-radius absolute residual profiles and the corresponding leave-one-galaxy-out threshold. This visualization is not a primary endpoint and is not used to tune the classifier.

5 Baseline-family comparison

The baseline comparison weakens any projection-specific claim and shows that the marginal association is shared across low-acceleration residual families. MOND-simple [7] and empirical RAR-like [6] residual scores also separate A/C systems, while the Newtonian baryonic RMS score is near chance. The defensible conclusion is that A/C separation is concentrated in low-acceleration residual-family scores, not that the projection family owns unique information.

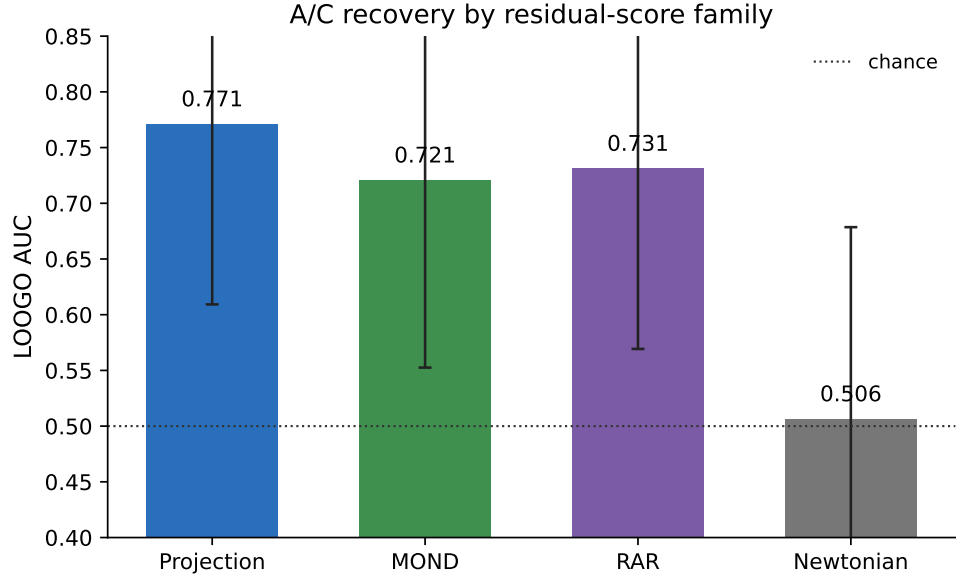


Figure 3: Held-out A/C recovery across residual-score families. The Newtonian baryonic RMS score is near chance.

This pattern is compatible with several physical readings: disturbed systems may violate smooth equilibrium assumptions, non-circular motions may be more important in low-acceleration regions, or observability may expose more disturbance structure in particular subsets. The present data do not distinguish these explanations.

6 Failure modes

The error audit is retained because a useful diagnostic must show where it fails. False-negative C systems demonstrate that externally disturbed galaxies do not always produce large residual burden under a smooth rotation-curve score. False-positive A systems show that residual structure can arise without a strong external disturbance label.

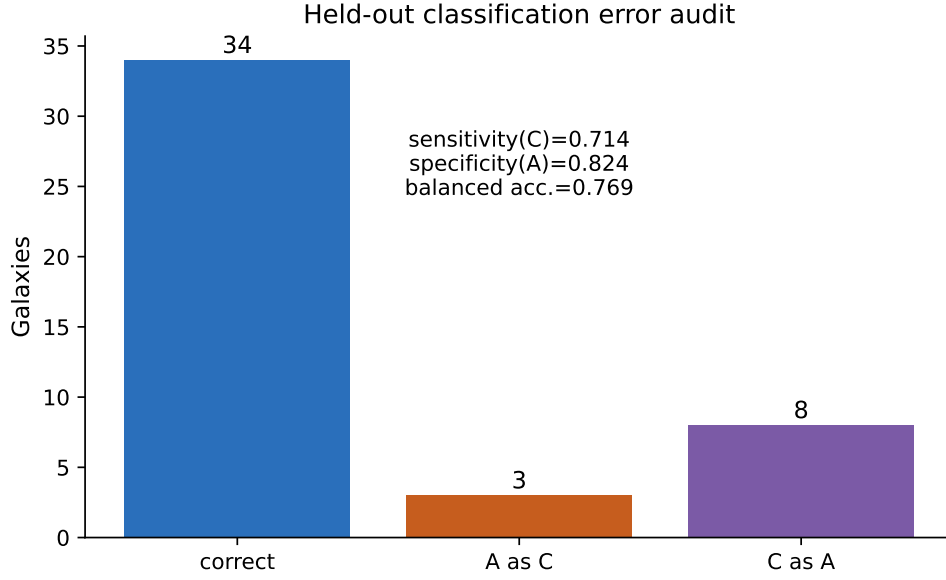


Figure 4: Held-out classification outcome counts for the Projection RMS threshold rule.

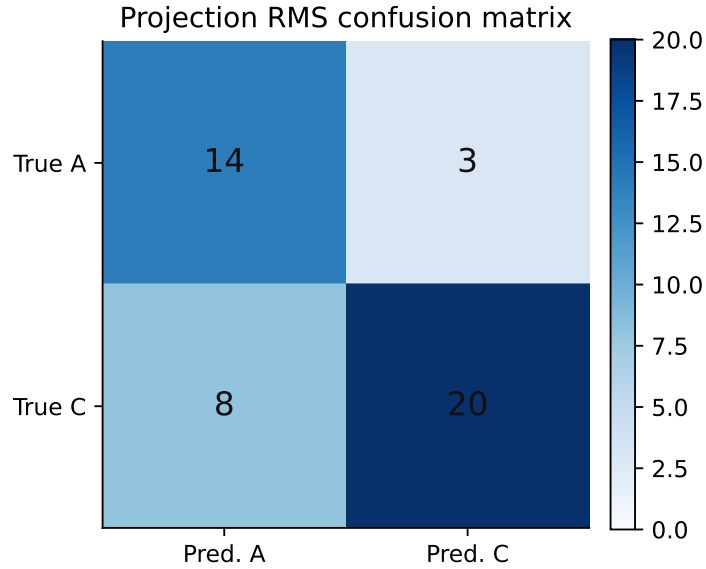


Figure 5: Confusion matrix for the held-out Projection RMS rule.

These failures are not relabeling evidence. They are targets for follow-up inspection, especially where residual morphology, inclination, radial coverage, and HI kinematic asymmetry disagree.

CamB is the most important failure case. It is externally labeled A from regular-kinematics evidence, but it has Projection RMS near 0.96 and only nine rotation-curve points. The paper therefore treats CamB as a named residual-hard outlier rather than letting it disappear inside summary statistics.

Table 3: B-class sensitivity check. B systems are not used as primary labels.

N_B	Threshold	B C-like	B A-like	Interpretation
28	0.166489797	13	15	descriptive only

Table 4: Named outliers and failure-case inspection.

Galaxy	Class	Error	Score	Points	Distance	Incl.	Inspection note
CamB	A	A as C	0.963585384	9	3.36	65.0	dominant A-class failure; low point count; very high low-acceleration residual burden
UGC05764	A	A as C	0.329468364	10	7.47	60.0	secondary A-class failure; compact 10-point curve; high residual margin
NGC5585	A	A as C	0.173858350	24	7.06	51.0	near-threshold A-class failure; external low-asymmetry evidence but residual just above gate

7 External proxy readouts

The external HI context draws on WHISP lopsidedness and morphology [12, 13, 2], THINGS non-circular-motion diagnostics [11], Reynolds et al. HI asymmetry catalogues [10], and ALFALFA profile asymmetry [14]. Table 5 lists each frozen proxy gate. WHISP resolved-HI asymmetry is the strongest supportive external readout, with AUC=0.714 at small overlap. Reynolds/LVHIS is also directionally strong (AUC=0.778) but below the frozen $N \geq 15$ gate. WHISP morphology is mixed, and ALFALFA/HALOGAS do not provide strong directional support. The external evidence therefore adds real context for the residual-inference result, while still falling short of independent validation.

The strongest remaining weakness is external validation. All rotation-curve residual endpoints in this manuscript are SPARC-internal, and the auxiliary HI proxy readouts are contextual rather than a held-out replication. A fair skeptical reading is therefore: the association is reproducible within this SPARC packet, but it is not yet externally established.

Table 5: External-proxy readouts and frozen gates.

Source	Overlap	Direction/status	Effect size	Gate
WHISP resolved	14	positive, small N	r=0.391; AUC=0.714	context
WHISP morph.	25	mixed	AUC=0.644; burden=0.506	context
Reynolds/LVHIS	6	promising, N<15	r=0.375; AUC=0.778	fail N
ALFALFA	22	weak	r=0.146; AUC=0.472	weak
HALOGAS	5	control-like	r=0.216; AUC=0.500	control
THINGS route2	0_score_ready	closed	0 score-ready	negative audit

8 External-validation status

The present manuscript should not be read as a completed external-validation study. It contains a positive marginal SPARC discrimination result, small-overlap external-proxy consistency checks, a negative THINGS expansion audit, and the downstream negative conditional/proxy results recorded below. Together these establish a reproducible unconditional association and define the next test, but they do not establish projection-specific conditional information or replace an independent matched-tracer replication.

A paper-grade Phase II replication must be a source-family holdout. The evidence rule, disturbance labels, observability covariates, minimum overlap, and pass/fail thresholds must be frozen before any endpoint readout. The minimum acceptable target remains $N \geq 15$ score-ready galaxies in a non-SPARC source family or in a demonstrably independent kinematic-proxy family. If the effect disappears, reverses, or becomes dominated by distance, inclination, point count, or beam-size proxies, the correct conclusion is that the current result is SPARC-specific, proxy-specific, or observability-driven.

9 Downstream conditional and morphology-proxy audits

A later Paper 3 fixed-model audit tested predictive increment on the same frozen 45-galaxy A/C sample using repeated stratified five-fold cross-fitting with 20 repetitions, endpoint-residual-free observability/baryonic baselines, and residual structure common to MOND/RAR-like comparators. The unique projection contrast failed to improve held-out proper scores under either declared baseline and any of the three declared regularization values. At the primary O7B7, $C = 0.3$ setting, the AUC change was -0.01890756 , the log-loss improvement was -0.03994321 , and the Brier-score improvement was -0.01188683 ; negative improvement means worse performance. This is a fixed-model conditional predictive-increment audit, not a literal conditional-mutual-information estimate and not proof that every possible additional rotation signal is absent.

A separate source-frozen seven-galaxy EDGE-CALIFA rotation-morphology proxy endpoint returned mean $D = -0.058679293503004035$, exact one-sided $p = 0.6015625$, median $D = +0.11020194395988532$, and 4/7 positive galaxy-level values. It failed its directional prevalidation gate. This is an external morphology-proxy stress test, not an independent Tau-specific test and not a direct replication of the Paper 2 A/C classifier.

Further Paper 8 public-data routes also remain non-promotable. A prospective LITTLE THINGS transfer test contains 14 galaxies and 313 points but only one frozen morphology family; it improves mean RMSE over TPG/v6 by 0.101 km s^{-1} and is worse than MOND by 0.649 km s^{-1} [8]. In PHANGS [3, 1], a three-galaxy low-order morphology-orthogonal statistic is null ($p = 0.8082$). A separate two-galaxy $m = 2$ statistic is numerically non-null, but the $m = 1$ control is also non-null and the source-label integrity audit fails. A higher-dimensional four-galaxy confirmatory packet stops at its predeclared sector-occupancy gate without releasing a score. These are distinct proxy and transfer tests, not direct repeats of the Paper 2 classifier.

None of these downstream audits changes the original AUC, shuffle, or bootstrap calculations. They change the interpretation: the internal result is a marginal or unconditional association, while conditional projection specificity and independent external replication remain unestablished.

They also constrain only the finite morphology representations that were actually frozen. If a representation Φ_k maps two possible body states to the same source record while their complete terminal images differ, no terminal rule can factor through Φ_k ; failure then rejects or limits that representation rather than every possible richer body. The converse leakage guard is essential: after endpoint y is inspected, augmenting the source span by y makes its projected residual vanish identically. A refined representation must therefore be source-only, frozen in advance, and evaluated on a new untouched packet.

10 THINGS route2 negative audit

The THINGS expansion route was audited because it could have raised the independent-control overlap. It is now closed as not score-ready. The required machine-readable per-radius mass-

model columns were not recovered for the missing galaxies, and the reconstruction route failed its photometry and solver-validation gates before any missing-galaxy scores were computed.

This negative audit should remain visible. It documents why the paper does not claim THINGS $N \geq 15$, avoids plot-digitized or synthetic mass models, and prevents endpoint-driven data recovery from entering the evidence chain.

11 Observability and systematics

Observability remains the main scientific caveat. Nearby galaxies can reveal asymmetry and morphological disturbance more easily than distant systems; inclination, beam smearing, asymmetric drift, pressure support, and non-circular motion can all alter residual morphology. The current controls reduce but do not eliminate these concerns.

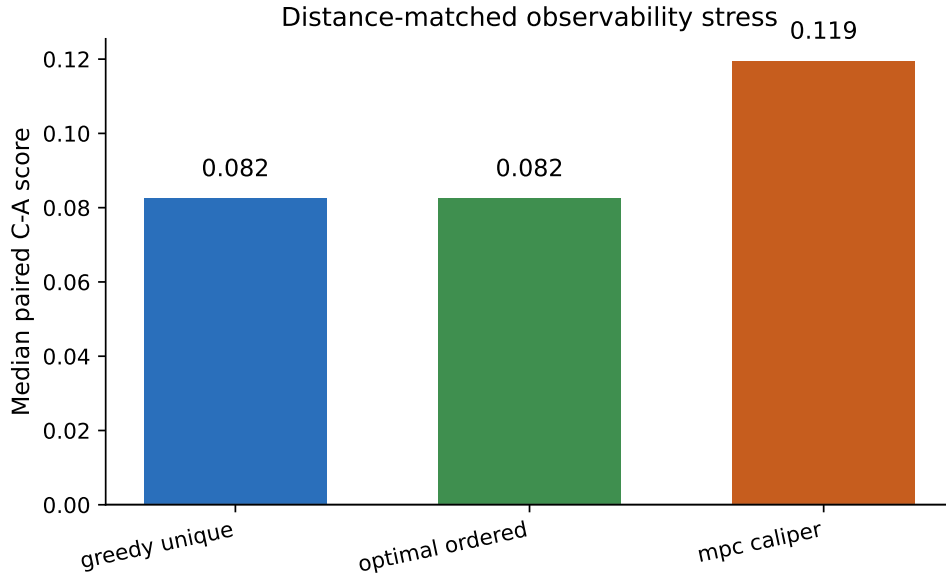


Figure 6: Distance-matched stress checks for the primary residual score. These checks reduce but do not eliminate observability concerns.

Accordingly, the result should be read as a reproducible marginal SPARC diagnostic association with mixed external-proxy context, not as a selection-function-proof physical inference.

The distance-stress labels are operational. Greedy unique matching pairs each A galaxy with one nearby C galaxy without reusing controls. Optimal ordered matching sorts both classes by distance and pairs by rank. The Mpc-caliper check keeps only pairs within the frozen maximum distance separation. These checks are not a full selection-function model.

Table 6: Observability and nuisance-covariate stress summaries.

Check	N	Value	Secondary value
partial distance residual correlation	45	0.434416486	nuisance controls
partial distance residual AUC	45	0.662000000	residual split
nuisance-only model R2	45	0.207747757	nuisance-only
point-count correlation	45	-0.256276232	nuisance channel
velocity-error correlation	45	-0.196321724	nuisance channel
inclination-error correlation	45	0.251685097	nuisance channel
distance-error correlation	45	-0.155582292	nuisance channel

This appendix-style stress table is not a full hierarchical model or a claim of bias removal. It documents that observability channels were inspected and that residual association should remain caveated until a larger external holdout can support multivariable inference.

12 Claim boundary and Phase II

Allowed claim: fixed residual-shape features show marginal or unconditional separation of externally reviewed A/C disturbance classes in the current SPARC packet; the separation is concentrated in low-acceleration residual families rather than Newtonian baryonic RMS, while external proxy readouts are mixed. The later conditional projection-contrast and EDGE-CALIFA proxy gates are negative; LITTLE THINGS is a caveated one-family transfer, and the current PHANGS routes fail null, specificity, label-integrity, or support gates.

Forbidden claims: Tau Core validation, gravity-model selection, projection-formula uniqueness, stable conditional projection-specific information, replacement of external labels by residual-only labels, broad independent external validation, EDGE-CALIFA as an independent Tau-specific test, or THINGS route2 positive evidence.

The later Tau Core full-4D descent changes neither the frozen feature map nor any AUC, permutation, or bootstrap result in this paper. It shows only that a terminal velocity residual can in principle have body, lapse, shift, spatial-metric, photon-transfer, astrophysical, and observational contributions. The present classifier does not reconstruct those contributions, and no nonzero radial Tau coframe may be selected from its labels or residuals. The later exact compiler also proves that $C \neq 0$ in a parent Hessian need not produce any Tau-over-standard excess: the scoreable quantity would be $E_K = (K_{HH} - K_{\text{std}}) - CK_V^{-1}C^\dagger$. It is not reconstructed by the residual-shape classifier.

The next paper-grade step is a new source-frozen independent matched-tracer or external source-family test with $N \geq 15$, no velocity-endpoint refit, explicit observability and baryonic covariates, declared standard/common residual comparators, and predefined failure conditions. A negative Phase II result should be treated as evidence that the present association is SPARC-specific, proxy-specific, or observability-driven.

13 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer–source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer Q_{OS} .

Ownership and use. The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

Consequence for this paper. For galactic inference, these observer constructions do not derive a rotation-curve correction or identify a measured residual as a parent effect. Existing endpoint freezes and scores are unchanged.

14 Conclusion

This paper finds marginal separation of externally reviewed A/C disturbance classes by fixed residual-shape features within the current SPARC packet. The strongest internal endpoint is Projection RMS, with LOOGO AUC=0.771008403, shuffled-label $p = 0.002000000$, and a bootstrap distribution centered near the observed value. MOND-simple and empirical RAR-like residual scores also separate the classes, while the Newtonian baryonic RMS control is near chance. The result is therefore an unconditional low-acceleration residual-family association with structural disturbance, not a projection-formula-specific detection.

The paper does not establish a new gravity model, does not validate Tau Core, and does not replace external disturbance labels with residual-only labels. The later Paper 3 conditional projection-contrast audit and the seven-galaxy EDGE–CALIFA morphology-proxy stress test are negative; the LITTLE THINGS and PHANGS continuations likewise do not pass a claim-raising external gate. The paper therefore does not establish conditional projection-specific information or

completed external validation. CamB and the other named failure cases show that residual burden can be high in externally regular systems, and the observability checks show that distance, radial coverage, inclination, point count, and HI data quality remain live alternative explanations.

The next decisive test is a new source-frozen independent matched-tracer or external source-family replication with at least $N \geq 15$ score-ready galaxies, observability and baryonic covariates, common residual comparators, and predefined failure conditions. If that test supplies held-out proper-score increment in the same direction, the present result becomes a stronger phenomenological claim. If it fails, the appropriate interpretation is that the association is SPARC-specific, proxy-specific, or observability-driven.

15 ROC appendix

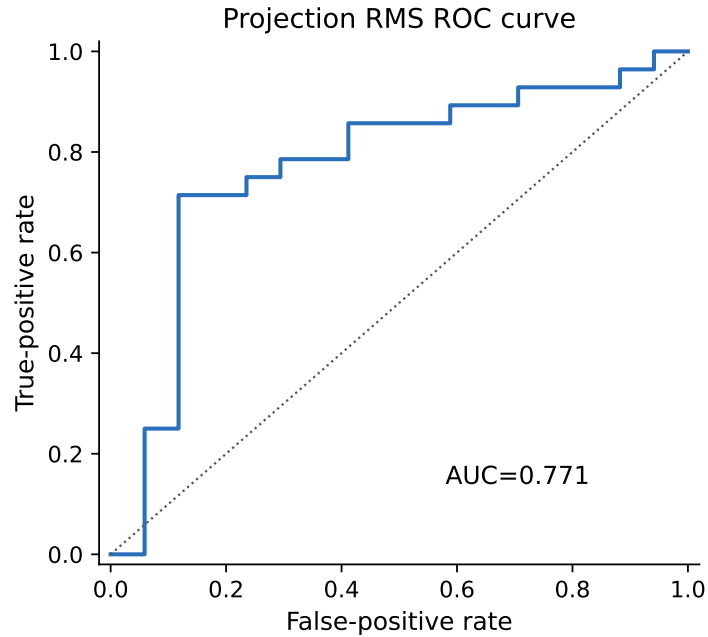


Figure 7: ROC curve for the Projection RMS score using the frozen A/C labels.

16 Stability and effect-size appendix

Table 7: Rank and stability effect-size summaries for Projection RMS.

Metric	Value	Secondary value
Mann-Whitney U	367.000000000	nA=17;nC=28
Cliff's δ	0.542016807	positive means C scores tend higher
AUC minus chance	0.271008403	auc=0.771008403
Bootstrap AUC median	0.775210084	p05=0.630252101;p95=0.894957983
Label-shuffle AUC median	0.497899160	p95=0.634453782;observed=0.771008403
AUC without CamB	0.819196429	A-class outlier removed for sensitivity only

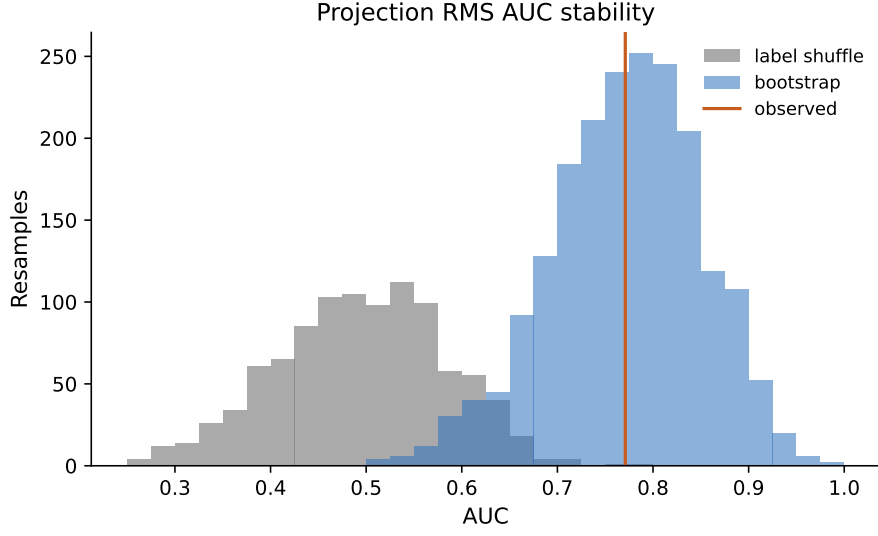


Figure 8: Bootstrap and shuffled-label AUC distributions for the Projection RMS endpoint.

A A/C sample table

Table 8: Frozen A/C sample used for the primary diagnostic.

Galaxy	Class	Distance	Incl.	Points	Projection RMS
CamB	A	3.36	65.0	9	0.963585384
DDO154	A	4.04	64.0	12	0.138812041
ESO079-G014	A	28.70	79.0	15	0.116776650
ESO116-G012	A	13.00	74.0	15	0.118483994
IC2574	C	3.91	75.0	34	0.221285260
NGC0055	C	2.11	77.0	21	0.204516651
NGC1705	C	5.73	80.0	14	0.354417125
NGC2366	C	3.27	68.0	26	0.250878781
NGC2403	A	3.16	63.0	73	0.084927422
NGC2976	C	3.58	61.0	27	0.239111203
NGC3109	C	1.33	70.0	25	0.175872342
NGC3198	A	13.80	73.0	43	0.156224718
NGC3741	C	3.21	70.0	21	0.204417844
NGC3917	A	18.00	79.0	17	0.143599282
NGC3972	A	18.00	77.0	10	0.101666153
NGC4183	A	18.00	82.0	23	0.067793794
NGC4559	A	9.00	67.0	32	0.133934337
NGC5055	C	9.90	55.0	28	0.117536453
NGC5585	A	7.06	51.0	24	0.173858350
NGC6503	A	6.26	74.0	31	0.072498887
NGC7331	A	14.70	75.0	36	0.098042642
UGC00128	A	64.50	57.0	22	0.138644763
UGC00191	A	17.10	45.0	9	0.124355499

Galaxy	Class	Distance	Incl.	Points	Projection RMS
UGC00731	C	12.50	57.0	12	0.339703437
UGC02455	C	6.92	51.0	8	0.793075490
UGC03580	C	20.70	63.0	47	0.280295930
UGC04305	C	3.45	40.0	22	0.595364038
UGC04325	C	9.60	41.0	8	0.284250584
UGC04499	C	12.50	50.0	9	0.137445134
UGC05253	C	22.90	37.0	73	0.080485995
UGC05716	A	21.30	54.0	12	0.113234582
UGC05764	A	7.47	60.0	10	0.329468364
UGC05829	C	8.64	34.0	11	0.138511849
UGC05918	C	7.66	46.0	8	0.142410680
UGC06446	C	12.00	51.0	17	0.194477890
UGC06818	C	18.00	75.0	8	0.308337913
UGC06917	C	18.00	56.0	11	0.071490152
UGC06973	C	18.00	71.0	9	0.212731540
UGC07323	C	8.00	47.0	10	0.146477589
UGC07399	C	8.43	55.0	10	0.397297780
UGC07577	C	2.59	63.0	9	0.654467132
UGC07603	C	4.70	78.0	12	0.180772538
UGC08490	C	4.65	50.0	30	0.186480775
UGC08837	C	7.21	80.0	8	0.439660106
UGC12632	C	9.77	46.0	15	0.105256872

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